

Optical Density Monitoring

This protocol describes the steps to preparing and monitoring a liquid culture of *E. coli*. In genetic engineering, bacterial chassis are critical players due to their well-explored methods of genetic manipulation (i.e. DNA transformation). By preparing and maintaining healthy *E. coli* cultures, this method supports the groundwork.

Reagents

✓	Name	Quantity	Location/Notes
☐	Liquid culture of <i>E. coli</i>	1	
☐	Liquid LB media	1mL	Blank

Equipment

✓	Name	Quantity	Location/Notes
☐	Spectrophotometer		
☐	Disposable cuvettes	4	(per reading)
☐	Vortex mixer		

Instructions

1. Aliquot 1mL of LB media into a cuvette, this is the blank.
2. Vortex stock bacteria to create a homogenized solution, vortex at 6 (look at notes for context)
3. Aliquot 1 mL of stock bacteria into a cuvette. Repeat 2x. You should have 3 cuvettes of stock bacteria and 1 cuvette of LB medium.

-
4. Turn on the spectrophotometer, set to the OD program, and set the wavelength to 600 nm.
 5. Use the blank sample to calibrate the system.
 6. The OD is the average of the 3 samples.

Notes here

- Growth stages:
 - Lag = cells adjusting to environment; not growing yet
 - Log = fast growth (doubling every ~20-30 minutes)
 - Stationary phase (saturation) = cell division slows/stops due to nutrients running out, limited oxygen, and waste buildup
- Mid-log phase occurs between OD₆₀₀ = 0.4-0.9 (best working range; fast growth)

Note on vortexing bacteria: Usually a vortex ranges from 0-10 with intensity, to avoid cell wall damage to the bacteria (and most cells you work with), vortex at a level of 6. If not an option, pulse the sample (gently press down and bring it back up straight away. Repeat a couple times.

References

- Adapted from E. coli Culture Preparation and OD monitoring (Synbio 2025) and  Bacteria (E. coli) Culture Conditions

Clark, B. (2022, June 28). *E. coli* Culture. protocols.io.

[dx.doi.org/10.17504/protocols.io.5jyl8999o7v2w/v1](https://doi.org/10.17504/protocols.io.5jyl8999o7v2w/v1)

Islam, P. (2018, Nov 23). *Optical density for Bacterial solution (Worm Food)*.

protocols.io. [dx.doi.org/10.17504/protocols.io.vude6s6](https://doi.org/10.17504/protocols.io.vude6s6)